23CS10012

Assignment 2: Opinion Maximisation on ego-Facebook

Reproducible numerical implementation and experiments. All target IDs use original SNAP labels.

Software: {"dataset_sha256": "f41c026ed8af3cc3359f1ca5573d0605fb09ae0eefa34544b820fd8c6e2ef296", "nodes": 4039, "edges": 88234, "components": 1, "raw_lines": 88234, "removed_loops": 0, "collapsed_duplicates": 0, "seed": 2026, "streams": {"alpha": 0, "U": 1, "P": 2, "random": "1000 + pool_code*100 + trial", "pgd": 3000}, "python": "3.12.14", "numpy": "2.5.4", "scipy": "1.18.1", "networkx": "3.6.1", "pandas": "3.1.1", "matplotlib": "3.11.1", "machine": "x86_64", "processor": "", "cpus": 5, "threads": {"OPENBLAS_NUM_THREADS": "1", "OMP_NUM_THREADS": null, "MKL_NUM_THREADS": null}}

Execution hardware: {"cpu_model": "INTEL(R) XEON(R) PLATINUM 8573C", "available_cpus": 5, "affinity_cpus": 5, "cpu_quota": "400000 100000", "memory_max": "34359738368", "threads_per_worker": 1}

Data: official SNAP facebook_combined.txt, undirected simple graph; input SHA-256 f41c026ed8af3cc3359f1ca5573d0605fb09ae0eefa34544b820fd8c6e2ef296. Duplicate edges collapse and self loops are removed; observed counts: {'nodes': 4039, 'edges': 88234, 'components': 1, 'removed_loops': 0, 'collapsed_duplicates': 0}

Bounds [0.001,1]; budgets 1,2,3,4,5,10,20,30. Initial alpha uniform; U uniform; P Beta(0.5,0.5). SeedSequence([2026,stream]) with alpha=0,U=1,P=2. The same alpha is shared across profiles. Random stream=1000+100*pool_code+trial (20 trials); PGD random start=3000.

Reproduction: python Assignment2_23CS10012_analysis_task4.py prepare; python Assignment2_23CS10012_analysis_task4.py run; python Assignment2_23CS10012_analysis_task4.py report. Use --dataset and --output to override paths. commands.json records each task command. requirements.txt pins tested versions. Set OPENBLAS_NUM_THREADS=OMP_NUM_THREADS=MKL_NUM_THREADS=1 before Python.

Timing includes algorithm initialization and numerical verification. Each main algorithm runs in a fresh Python subprocess. Peak memory is Linux total-process ru_maxrss / 1024 MiB, including native arrays. Random time is initial setup plus the sum of 20 trial prefix times; its shading is one sample standard deviation, not a confidence interval.

Main findings

U max: initial f=2034.815972; greedy at k=30 gives f=2087.060183, improvement 52.244210. Cumulative time 8.502 s, peak RSS 610.41 MiB.

P max: initial f=2006.049365; greedy at k=30 gives f=2091.745296, improvement 85.695931. Cumulative time 9.348 s, peak RSS 608.93 MiB.

U min: initial f=2034.815972; greedy at k=30 gives f=1980.341463, improvement 54.474510. Cumulative time 9.521 s, peak RSS 608.99 MiB.

U matching-shortlist ten rounds: fresh solves 50.469 s versus fast 5.085 s (9.92x speedup), with matching targets and endpoint assignments.

P matching-shortlist ten rounds: fresh solves 57.249 s versus fast 5.934 s (9.65x speedup), with matching targets and endpoint assignments.

These comparisons describe one recorded draw of the initial resistances and each opinion profile. Random variability is conditional on those fixed inputs; it is not uncertainty over graph samples or opinion-profile draws.

The required PGD experiments and an independent reference agree numerically for all three starts in all four profile/objective combinations. The false convexity rationale is explicitly separated from the valid endpoint theorem and unbudgeted solvability.

Restricted exact references: U: |C|=26; absolute greedy/reference discrepancies at k=2,3 are [9.094947017729282e-13, 1.8189894035458565e-12]; P: |C|=24; absolute greedy/reference discrepancies at k=2,3 are [1.318767317570746e-11, 1.0686562745831907e-11]. This agreement on small budgets does not imply a global approximation guarantee.

Bonus: 24 of 24 tested scenario/budget starts were already locally optimal for the specified flip/shortlist-swap neighborhood. Zero improvement is reported without claiming optimality outside that neighborhood.

Task 1: equilibrium and endpoint checks

M=I-(I-A)P, b=A s; solve Mz=b with checked bicgstab (rtol and atol 2e-13), falling back to a checked sparse direct solve on failure. No specialized opinion-dynamics package is used.

For two nodes, D=alpha_a+alpha_b-alpha_a*alpha_b. z_a=[alpha_a*s_a+(1-alpha_a)*alpha_b*s_b]/D; z_b is the symmetric expression. Checks: [{"alpha": [0.2, 0.8], "s": [0.1, 0.9], "exact": [0.7095238095238098, 0.8619047619047621], "computed": [0.7095238095238096, 0.861904761904762]}, {"alpha": [0.001, 1.0], "s": [1.0, 0.0], "exact": [0.0010000000000000002, 0.0], "computed": [0.001, 0.0]}, {"alpha": [1.0, 1.0], "s": [0.3, 0.7], "exact": [0.3, 0.7], "computed": [0.3, 0.7]}]

Facebook direct iteration versus independent linear solve: {"U": {"f": 2034.8159724468524, "iteration_error": 1.0061396160665481e-12, "steps": 51}, "P": {"f": 2006.0493646605369, "iteration_error": 1.4654388813539754e-12, "steps": 53}}

For fixed other coordinates, the Sherman-Morrison expression makes f linear-fractional in one resistance, with a positive denominator and a derivative of constant sign. Thus a maximum lies at an endpoint (or every value ties). Coordinate replacement yields an endpoint optimum. Three 21-point sweeps are shown below.

Fresh greedy evaluates both endpoints at every remaining candidate, retains previous assignments, and uses the 200 highest-degree candidates on the full graph. Each ten-round profile uses 3910 candidate solves; verification and fallback solves are recorded separately. Ties use objective tolerance 1e-9, then original ID and lower endpoint.

Task 1 checkpoints: U

{"k": 1, "added": 1684, "index": 1684, "alpha": 0.001, "f": 2038.6040132645603, "gain": 3.7880408177079516, "seconds": 5.250659205999909, "peak_mib": 111.2421875, "candidate_solves": 400, "total_solves": 401, "recovery_solves": 0}

{"k": 2, "added": 348, "index": 348, "alpha": 1.0, "f": 2042.2167010558778, "gain": 3.612687791317512, "seconds": 12.465233919999719, "peak_mib": 111.2421875, "candidate_solves": 798, "total_solves": 799, "recovery_solves": 0}

{"k": 3, "added": 414, "index": 414, "alpha": 1.0, "f": 2045.720239250226, "gain": 3.503538194348266, "seconds": 18.428727242999685, "peak_mib": 111.2421875, "candidate_solves": 1194, "total_solves": 1195, "recovery_solves": 0}

{"k": 4, "added": 3437, "index": 3437, "alpha": 0.001, "f": 2049.122410616691, "gain": 3.402171366464927, "seconds": 22.866403340999568, "peak_mib": 111.2421875, "candidate_solves": 1588, "total_solves": 1589, "recovery_solves": 0}

{"k": 5, "added": 1431, "index": 1431, "alpha": 0.001, "f": 2050.9673479209114, "gain": 1.844937304220366, "seconds": 27.2498575919999, "peak_mib": 111.2421875, "candidate_solves": 1980, "total_solves": 1981, "recovery_solves": 0}

{"k": 10, "added": 2007, "index": 2007, "alpha": 0.001, "f": 2057.9925779744212, "gain": 1.1622321351392202, "seconds": 50.468623320000006, "peak_mib": 111.2421875, "candidate_solves": 3910, "total_solves": 3911, "recovery_solves": 0}

Task 1 checkpoints: P

{"k": 1, "added": 107, "index": 107, "alpha": 0.001, "f": 2025.0002512534795, "gain": 18.950886592942652, "seconds": 7.301603806999992, "peak_mib": 111.2421875, "candidate_solves": 400, "total_solves": 401, "recovery_solves": 0}

{"k": 2, "added": 1684, "index": 1684, "alpha": 1.0, "f": 2032.7265434012731, "gain": 7.726292147793629, "seconds": 13.955990371000098, "peak_mib": 111.2421875, "candidate_solves": 798, "total_solves": 799, "recovery_solves": 0}

{"k": 3, "added": 3437, "index": 3437, "alpha": 0.001, "f": 2039.5144078197275, "gain": 6.787864418454319, "seconds": 21.63799204399993, "peak_mib": 111.2421875, "candidate_solves": 1194, "total_solves": 1195, "recovery_solves": 0}

{"k": 4, "added": 1912, "index": 1912, "alpha": 0.001, "f": 2043.8757175147566, "gain": 4.361309695029149, "seconds": 26.478416991999893, "peak_mib": 111.2421875, "candidate_solves": 1588, "total_solves": 1589, "recovery_solves": 0}

{"k": 5, "added": 0, "index": 0, "alpha": 0.001, "f": 2047.4591133897432, "gain": 3.583395874986536, "seconds": 30.34563996299994, "peak_mib": 111.2421875, "candidate_solves": 1980, "total_solves": 1981, "recovery_solves": 0}

{"k": 10, "added": 2333, "index": 2333, "alpha": 0.001, "f": 2056.502117231148, "gain": 1.2607441479540284, "seconds": 57.24862933099985, "peak_mib": 111.2421875, "candidate_solves": 3910, "total_solves": 3911, "recovery_solves": 0}

Task 2: fast greedy and baselines

Let G=M^-1, delta=alpha_new[v]-alpha[v], g=G[:,v], h=P[v,:]G, d=1+delta*h[v]. M_new=M+delta*e_v*P[v,:]. Sherman-Morrison gives G_new=G-delta*g*h/d. With the changed RHS, z_new=z+delta*(s[v]-(Pz)[v])*g/d; the gain is the same scalar times sum(g).

Cache c=1^T G and t=diag(PG). With beta=delta/d, c_new=c-beta*c[v]*h and t_new=t-beta*(P*g)*h. All candidate evaluations use cached scalars; no candidate equilibrium solve. Each accepted state is checked by an independent solver. No division by 1-alpha is used.

Both profiles match every Task 1 shortlist choice and objective through ten rounds within 1e-7; the driver aborts on a mismatch. All-node runs cover 30 rounds for U-max, P-max and U-min.

Baseline I ranks innate s. Baseline II ranks degree/(2m) times s/sum_neighbor(s). Both assign selected alpha=1, as in paper Section 5. Minimisation ranks 1-s but evaluates original s. Positive/zero is infinity and 0/0 is zero; ties use IDs.

Random samples 20 uniform permutations, using nested prefixes. At each step it tests both endpoints given earlier assignments and keeps the better one. It does not jointly reoptimize the selected set. Fixed-base Woodbury systems avoid new full solves. All-node and C-restricted universes are labeled separately.

Task 2: unbudgeted method and discrepancy

Write B=X-(X-I)P with x=1/alpha in [1,1000]. Differentiate Bz=s: df/dx_i=-q_i*((I-P)z)_i, where B^T q=1. PGD minimizes -f for maximisation and f for minimisation using projection, adaptive spectral trial steps, and Armijo backtracking. Initial points are 1/alpha, the box midpoint, and a recorded random interior point.

The assignment states this objective is convex. A two-node counterexample with s=(1,0) gives f(x,y)=(2y-1)/(x+y-1). f(2,2)=1 > [f(2,1)+f(2,3)]/2=7/8; it is also below [f(1,2)+f(3,2)]/2=9/8. We execute the requested PGD experiment but do not infer convexity or global optimality from matching runs.

An independent endpoint-policy reference is obtained by policy improvement and certified numerically using H_i(z)=max/min over a in {0.001,1} of a*s_i+(1-a)*(Pz)_i. H is monotone and a contraction with factor <=0.999. Its unique fixed point bounds all policy equilibria and is attained by its endpoint actions. For a freshly solved policy, n*||H(z)-z||_infinity/0.001 bounds total-objective error in exact arithmetic; reported floating-point residual certificates are numerical, not interval proofs.

Claim 1: the unbudgeted optimum does admit a polynomial-size linear-program formulation in equilibrium-value coordinates, although the claimed convexity in X is false. For maximisation, minimize sum(v) subject to v_i >= a*s_i+(1-a)*(Pv)_i for every i and both a in {l,u}. These inequalities say v>=H_max(v); monotonicity and contraction imply v>=v*, while v* is feasible and attained by its maximizing endpoint policy. For minimisation, maximize sum(v) with both inequalities reversed: every feasible v lies below the fixed point of H_min. Each LP has n variables and 2n linear inequalities, giving polynomial-time solvability for rational inputs under the standard LP bit-complexity model. This is an independent argument, not a claim that PGD has a global polynomial convergence guarantee. The experiments continue to use the assigned PGD method.

Claim 2: the coordinate endpoint theorem holds; PGD endpoint counts and independent reference assignments below show what these runs actually produce.

The complement identity f_(1-s)=n-f_s follows from B*1=1 and is tested numerically.

The corrected/superseding paper, Abebe et al. (2020), arXiv:2011.02583v1, acknowledges the earlier correctness issue in its abstract and introduction (Comparison with Previous Versions), proves marginal monotonicity in Section 5, and describes a polynomial local-search construction in Section 6. Our two-node X-coordinate counterexample directly addresses the substitution stated in this assignment.

Unbudgeted U max

Reference: {"f": 3904.478179057248, "bellman_residual": 5.014877402231832e-13, "objective_error_bound": 2.025508982761437e-06, "iterations": 8, "seconds": 0.1946615409997321}

{"f": 3904.478179057248, "iterations": 65, "status": "reference_certified", "endpoint_count": 4039, "seconds": 4.683696373000203, "peak_mib": 111.2421875, "start": 0, "reference_gap": 0.0}

{"f": 3904.478179057248, "iterations": 39, "status": "reference_certified", "endpoint_count": 4039, "seconds": 9.658690834000026, "peak_mib": 111.39453125, "start": 1, "reference_gap": 0.0}

{"f": 3904.478179057248, "iterations": 73, "status": "reference_certified", "endpoint_count": 4039, "seconds": 9.329537046000041, "peak_mib": 111.39453125, "start": 2, "reference_gap": 0.0}

Full alpha vectors and optimization histories are in the corresponding raw JSON file. Small-graph PGD versus exhaustive endpoint checks: [{"nodes": 6, "gradient_error": 2.3160535989052988e-11, "pgd": [{"direction": 1, "start": 0, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 19, "status": "reference_certified"}, {"direction": 1, "start": 1, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 10, "status": "reference_certified"}, {"direction": 1, "start": 2, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 22, "status": "reference_certified"}, {"direction": -1, "start": 0, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.662936703425657e-15, "iterations": 25, "status": "reference_certified"}, {"direction": -1, "start": 1, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.662936703425657e-15, "iterations": 26, "status": "reference_certified"}, {"direction": -1, "start": 2, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.662936703425657e-15, "iterations": 13, "status": "reference_certified"}]}, {"nodes": 5, "gradient_error": 4.2202019656656375e-11, "pgd": [{"direction": 1, "start": 0, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 23, "status": "reference_certified"}, {"direction": 1, "start": 1, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 8, "status": "reference_certified"}, {"direction": 1, "start": 2, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 14, "status": "reference_certified"}, {"direction": -1, "start": 0, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 20, "status": "reference_certified"}, {"direction": -1, "start": 1, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 23, "status": "reference_certified"}, {"direction": -1, "start": 2, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 20, "status": "reference_certified"}]}]

Unbudgeted U min

Reference: {"f": 132.53970168597408, "bellman_residual": 2.6305346789712303e-14, "objective_error_bound": 1.0624729568364799e-07, "iterations": 8, "seconds": 0.20215056300003198}

{"f": 132.53970168597408, "iterations": 52, "status": "reference_certified", "endpoint_count": 4039, "seconds": 3.6458386659996904, "peak_mib": 111.2421875, "start": 0, "reference_gap": -0.0}

{"f": 132.53970168597408, "iterations": 55, "status": "reference_certified", "endpoint_count": 4039, "seconds": 16.221639442000196, "peak_mib": 111.2421875, "start": 1, "reference_gap": -0.0}

{"f": 132.53970168597408, "iterations": 62, "status": "reference_certified", "endpoint_count": 4039, "seconds": 7.21138517099962, "peak_mib": 111.2421875, "start": 2, "reference_gap": -0.0}

Full alpha vectors and optimization histories are in the corresponding raw JSON file. Small-graph PGD versus exhaustive endpoint checks: [{"nodes": 6, "gradient_error": 2.3160535989052988e-11, "pgd": [{"direction": 1, "start": 0, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 19, "status": "reference_certified"}, {"direction": 1, "start": 1, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 10, "status": "reference_certified"}, {"direction": 1, "start": 2, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 22, "status": "reference_certified"}, {"direction": -1, "start": 0, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.6629367034256575e-15, "iterations": 25, "status": "reference_certified"}, {"direction": -1, "start": 1, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.6629367034256575e-15, "iterations": 26, "status": "reference_certified"}, {"direction": -1, "start": 2, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.6629367034256575e-15, "iterations": 13, "status": "reference_certified"}]}, {"nodes": 5, "gradient_error": 4.2202019656656375e-11, "pgd": [{"direction": 1, "start": 0, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 23, "status": "reference_certified"}, {"direction": 1, "start": 1, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 8, "status": "reference_certified"}, {"direction": 1, "start": 2, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 14, "status": "reference_certified"}, {"direction": -1, "start": 0, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 20, "status": "reference_certified"}, {"direction": -1, "start": 1, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 23, "status": "reference_certified"}, {"direction": -1, "start": 2, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 20, "status": "reference_certified"}]}]

Unbudgeted P max

Reference: {"f": 3994.1834064455907, "bellman_residual": 8.04800670550776e-13, "objective_error_bound": 3.250589908354584e-06, "iterations": 7, "seconds": 0.12778748399978213}

{"f": 3994.1834064455907, "iterations": 52, "status": "reference_certified", "endpoint_count": 4039, "seconds": 2.8265798520001226, "peak_mib": 111.2421875, "start": 0, "reference_gap": 0.0}

{"f": 3994.1834064455907, "iterations": 39, "status": "reference_certified", "endpoint_count": 4039, "seconds": 8.30163712500007, "peak_mib": 111.2421875, "start": 1, "reference_gap": 0.0}

{"f": 3994.1834064455907, "iterations": 68, "status": "reference_certified", "endpoint_count": 4039, "seconds": 7.440407811000114, "peak_mib": 111.2421875, "start": 2, "reference_gap": 0.0}

Full alpha vectors and optimization histories are in the corresponding raw JSON file. Small-graph PGD versus exhaustive endpoint checks: [{"nodes": 6, "gradient_error": 2.3160535989052988e-11, "pgd": [{"direction": 1, "start": 0, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 19, "status": "reference_certified"}, {"direction": 1, "start": 1, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 10, "status": "reference_certified"}, {"direction": 1, "start": 2, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 22, "status": "reference_certified"}, {"direction": -1, "start": 0, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.6629367034256575e-15, "iterations": 25, "status": "reference_certified"}, {"direction": -1, "start": 1, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.6629367034256575e-15, "iterations": 26, "status": "reference_certified"}, {"direction": -1, "start": 2, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.6629367034256575e-15, "iterations": 13, "status": "reference_certified"}]}, {"nodes": 5, "gradient_error": 4.2202019656656375e-11, "pgd": [{"direction": 1, "start": 0, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 23, "status": "reference_certified"}, {"direction": 1, "start": 1, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 8, "status": "reference_certified"}, {"direction": 1, "start": 2, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 14, "status": "reference_certified"}, {"direction": -1, "start": 0, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 20, "status": "reference_certified"}, {"direction": -1, "start": 1, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 23, "status": "reference_certified"}, {"direction": -1, "start": 2, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 20, "status": "reference_certified"}]}]

Unbudgeted P min

Reference: {"f": 47.75980877879526, "bellman_residual": 2.3647750424515834e-14, "objective_error_bound": 9.551326396461945e-08, "iterations": 8, "seconds": 0.14674492399990413}

{"f": 47.75980877879526, "iterations": 62, "status": "reference_certified", "endpoint_count": 4039, "seconds": 3.457965732000048, "peak_mib": 111.2421875, "start": 0, "reference_gap": -0.0}

{"f": 47.75980877879526, "iterations": 53, "status": "reference_certified", "endpoint_count": 4039, "seconds": 12.026482210999802, "peak_mib": 111.2421875, "start": 1, "reference_gap": -0.0}

{"f": 47.75980877879526, "iterations": 55, "status": "reference_certified", "endpoint_count": 4039, "seconds": 9.868933839999954, "peak_mib": 111.2421875, "start": 2, "reference_gap": -0.0}

Full alpha vectors and optimization histories are in the corresponding raw JSON file. Small-graph PGD versus exhaustive endpoint checks: [{"nodes": 6, "gradient_error": 2.3160535989052988e-11, "pgd": [{"direction": 1, "start": 0, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 19, "status": "reference_certified"}, {"direction": 1, "start": 1, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 10, "status": "reference_certified"}, {"direction": 1, "start": 2, "f": 5.693522714274982, "exact": 5.6935227142750024, "gap": 2.042810365310288e-14, "iterations": 22, "status": "reference_certified"}, {"direction": -1, "start": 0, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.6629367034256575e-15, "iterations": 25, "status": "reference_certified"}, {"direction": -1, "start": 1, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.6629367034256575e-15, "iterations": 26, "status": "reference_certified"}, {"direction": -1, "start": 2, "f": 1.1290611950905944, "exact": 1.1290611950905898, "gap": 4.6629367034256575e-15, "iterations": 13, "status": "reference_certified"}]}, {"nodes": 5, "gradient_error": 4.2202019656656375e-11, "pgd": [{"direction": 1, "start": 0, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 23, "status": "reference_certified"}, {"direction": 1, "start": 1, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 8, "status": "reference_certified"}, {"direction": 1, "start": 2, "f": 3.2388891453194923, "exact": 3.238889145319492, "gap": -4.440892098500626e-16, "iterations": 14, "status": "reference_certified"}, {"direction": -1, "start": 0, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 20, "status": "reference_certified"}, {"direction": -1, "start": 1, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 23, "status": "reference_certified"}, {"direction": -1, "start": 2, "f": 0.10808600817512165, "exact": 0.10808600817512115, "gap": 4.996003610813204e-16, "iterations": 20, "status": "reference_certified"}]}]

Task 2: non-submodularity and claim verdicts

The supplied published paper labels the requested example Example 4, not Example 13. Triangle s=(1,0,0), alpha=(0.1,0.1,0.1): {"sets": [[], [1], [3], [1, 2], [1, 3], [1, 2, 3]], "f": [1.0000000000000004, 2.6363636363636367, 1.4927824788451978, 2.8047018607501046, 2.804701860750105, 2.9960039960039966], "nonsubmodular": {"small": 0.1683382243864684, "large": 0.19130213525389195}, "nonsupermodular": {"small": 0.4927824788451973, "large": 0.1683382243864684}}

The paper prints 1.493 for f({3})-f(empty), omitting f(empty)=1. The corrected marginal is about 0.493 and the non-supermodularity violation remains.

Claim 3 verdict by profile (both distinct inequalities witnessed): {"U": true, "P": true}

Each Facebook witness evaluates the optimized set function by exhaustive endpoint assignments at all four sets and verifies those assignments with fresh solves. Larger marginal at B violates submodularity; smaller marginal at B violates supermodularity.

U: {"witnesses": {"nonsubmodular": {"numerical_error_bound": 1.314955319631217e-05, "A": [2593], "B": [2593, 1048], "v": 959, "sets": [[2593], [959, 2593], [1048, 2593], [959, 1048, 2593]], "assignments": [[1.0], [1.0, 1.0], [0.001, 1.0], [1.0, 0.001, 1.0]], "f": [2035.3221459736797, 2035.6416070757998, 2035.4126097855328, 2035.7323554416894], "marginal_small": 0.3194611021201581, "marginal_large": 0.3197456561565559, "attempt": 66}, "nonsupermodular": {"numerical_error_bound": 1.3495284306719313e-05, "A": [2496], "B": [2496, 2399], "v": 2194, "sets": [[2496], [2194, 2496], [2399, 2496], [2194, 2399, 2496]], "assignments": [[1.0], [1.0, 1.0], [1.0, 1.0], [1.0, 1.0, 1.0]], "f": [2034.8886893179329, 2035.4053014864078, 2035.0257616872407, 2035.5368510729709], "marginal_small": 0.5166121684749214, "marginal_large": 0.5110893857302017, "attempt": 98}}, "attempts": 98, "complete": true}

P: {"witnesses": {"nonsubmodular": {"numerical_error_bound": 2.0659003313605595e-05, "A": [], "B": [2614], "v": 2621, "sets": [[], [2621], [2614], [2614, 2621]], "assignments": [[], [0.001], [1.0], [1.0, 0.001]], "f": [2006.0493646605369, 2006.1177818048056, 2006.8050373424048, 2006.873757715332], "marginal_small": 0.06841714426877843, "marginal_large": 0.06872037292714595, "attempt": 71}, "nonsupermodular": {"numerical_error_bound": 2.0397126571092272e-05, "A": [2412], "B": [2412, 1645], "v": 1740, "sets": [[2412], [1740, 2412], [1645, 2412], [1645, 1740, 2412]], "assignments": [[1.0], [0.001, 1.0], [1.0, 1.0], [1.0, 0.001, 1.0]], "f": [2006.2768153075262, 2007.1912030162857, 2006.3415482267405, 2007.2515787667812], "marginal_small": 0.9143877087594774, "marginal_large": 0.9100305400406796, "attempt": 112}}, "attempts": 112, "complete": true}

Claim 4 verdict by scenario (strictly superior at EVERY requested budget): {"U_max": true, "P_max": true, "U_min": true}

Claim 4: signed greedy-minus-baseline differences at every requested budget (positive means strictly better): {"U_max": {"1": {"I": 3.503644903227041, "II": 3.4016912024310386}, "2": {"I": 6.374009159375646, "II": 6.3720705774746875}, "3": {"I": 8.888811452744903, "II": 9.65754849390055}, "4": {"I": 11.284061305004116, "II": 12.851557157742718}, "5": {"I": 13.570180552913826, "II": 15.464933534075499}, "10": {"I": 20.641456065672628, "II": 24.145338857885235}, "20": {"I": 29.14837881580138, "II": 36.35182784676135}, "30": {"I": 36.15796201985313, "II": 45.01488624140029}}, "P_max": {"1": {"I": 18.258413886406743, "II": 18.506526104919885}, "2": {"I": 25.230504083539017, "II": 25.526944417810455}, "3": {"I": 31.518462914790916, "II": 32.133745651369736}, "4": {"I": 35.568856442508604, "II": 36.77626065434242}, "5": {"I":

Task 2: non-submodularity and claim verdicts (continued)

39.81718457765692, "II": 41.13707348909452}, "10": {"I": 51.14114862001452, "II": 53.09822102934936}, "20": {"I": 62.15650359148685, "II": 68.60843468021721}, "30": {"I": 70.90109362291992, "II": 81.07807538429051}}, "U_min": {"1": {"I": 11.00419472133899, "II": 10.984120590305793}, "2": {"I": 14.98026596485306, "II": 15.045907924660014}, "3": {"I": 17.255389777206346, "II": 17.19908331304873}, "4": {"I": 19.62776241387246, "II": 19.512919602411102}, "5": {"I": 21.092248260004908, "II": 21.112397647168564}, "10": {"I": 26.380005206299984, "II": 27.31232810633469}, "20": {"I": 33.882793688021366, "II": 38.14685116178907}, "30": {"I": 40.87098659537378, "II": 45.415027327548614}}}

Do not generalize empirical superiority into an approximation guarantee.

Task 3 exact restricted reference: U

{"initial": 2034.8159724468514, "C_indices": [0, 107, 348, 376, 414, 428, 475, 483, 563, 698, 1211, 1352, 1431, 1621, 1663, 1684, 1730, 1800, 1888, 1912, 2266, 2313, 2347, 2543, 3437, 3980], "C_ids": [0, 107, 348, 376, 414, 428, 475, 483, 563, 698, 1211, 1352, 1431, 1621, 1663, 1684, 1730, 1800, 1888, 1912, 2266, 2313, 2347, 2543, 3437, 3980], "top_degree": [107, 1684, 1912, 3437, 0, 2543, 2347, 1888, 1800, 1663, 1352, 2266, 483, 348, 1730], "top_gain": [1684, 348, 414, 3437, 698, 3980, 1431, 428, 1621, 1211, 2543, 2313, 563, 376, 475], "overlap": 4, "pool_size": 26, "setup_seconds": 3.917754124000112, "references": {"2": {"f": 2042.2167010558778, "indices": [348, 1684], "ids": [348, 1684], "alpha": [1.0, 0.001], "configurations": 1300, "subsets": 325, "competitive_verified": 1, "estimated_runner_gap": 0.05033163058919854, "seconds": 0.11644404600019698, "total_seconds": 4.034200548000172, "peak_mib": 606.96484375}, "3": {"f": 2045.720239250226, "indices": [348, 414, 1684], "ids": [348, 414, 1684], "alpha": [1.0, 1.0, 0.001], "configurations": 20800, "subsets": 2600, "competitive_verified": 1, "estimated_runner_gap": 0.10143948825862026, "seconds": 1.4201342999999724, "total_seconds": 5.454381442999875, "peak_mib": 606.96484375}}, "complete": true}

For changed set S, let E select its columns, D=diag(delta), R=P[S,:]. Solve (I+D*R*G*E)w=D*(s[S]-R*z). Then f_new=f_base+(1^T*G*E)w. Enumerate all subsets and endpoint assignments; verify competitive configurations independently. Numerical ties use IDs and endpoints.

T* is optimal within C to the stated numerical tolerance, not necessarily over all 4039 nodes. Hence all-node gain ratios may exceed one. Jaccard is measured against the canonical reported reference; Random averages per-trial overlaps.

Task 3 exact restricted reference: P

{"initial": 2006.0493646605503, "C_indices": [0, 23, 107, 348, 414, 483, 698, 1070, 1352, 1431, 1663, 1684, 1730, 1800, 1888, 1912, 2266, 2347, 2543, 3366, 3397, 3437, 3830, 3980], "C_ids": [0, 23, 107, 348, 414, 483, 698, 1070, 1352, 1431, 1663, 1684, 1730, 1800, 1888, 1912, 2266, 2347, 2543, 3366, 3397, 3437, 3830, 3980], "top_degree": [107, 1684, 1912, 3437, 0, 2543, 2347, 1888, 1800, 1663, 1352, 2266, 483, 348, 1730], "top_gain": [107, 1684, 3437, 3980, 1912, 0, 698, 414, 23, 1431, 3830, 1070, 3397, 1800, 3366], "overlap": 6, "pool_size": 24, "setup_seconds": 5.647410157000195, "references": {"2": {"f": 2032.7265434012731, "indices": [107, 1684], "ids": [107, 1684], "alpha": [0.001, 1.0], "configurations": 1104, "subsets": 276, "competitive_verified": 1, "estimated_runner_gap": 0.9384293568152771, "seconds": 0.0735344240001723, "total_seconds": 5.720946981999987, "peak_mib": 607.52734375}, "3": {"f": 2039.5144078197275, "indices": [107, 1684, 3437], "ids": [107, 1684, 3437], "alpha": [0.001, 1.0, 0.001], "configurations": 16192, "subsets": 2024, "competitive_verified": 1, "estimated_runner_gap": 1.9068847158812332, "seconds": 1.0188431990000026, "total_seconds": 6.739879605000169, "peak_mib": 607.52734375}}, "complete": true}

For changed set S, let E select its columns, D=diag(delta), R=P[S,:]. Solve (I+D*R*G*E)w=D*(s[S]-R*z). Then f_new=f_base+(1^T*G*E)w. Enumerate all subsets and endpoint assignments; verify competitive configurations independently. Numerical ties use IDs and endpoints.

T* is optimal within C to the stated numerical tolerance, not necessarily over all 4039 nodes. Hence all-node gain ratios may exceed one. Jaccard is measured against the canonical reported reference; Random averages per-trial overlaps.

Task 4 comparison table

profile	direction	method	k	f	gain_fraction	seconds	peak_mib
U	max	fast	5	2051.966521	0.009173	4.805765	610.414062
U	max	fast	10	2061.520692	0.014283	5.476343	610.414062
U	max	fast	30	2087.060183	0.027943	8.501638	610.414062
U	max	I	5	2038.396340	0.001915	0.106691	111.242188
U	max	I	10	2040.879236	0.003243	0.184682	111.242188
U	max	I	30	2050.902221	0.008604	0.421150	111.242188
U	max	II	5	2036.501587	0.000902	0.093630	111.242188
U	max	II	10	2037.375353	0.001369	0.193478	111.242188
U	max	II	30	2042.045296	0.003867	0.695354	111.242188
U	max	random	5	2036.128366	0.000702	4.669085	608.453125
U	max	random	10	2037.303267	0.001330	4.687629	608.453125
U	max	random	30	2042.575947	0.004150	5.047124	608.453125
P	max	fast	5	2048.756722	0.021481	5.329656	608.933594
P	max	fast	10	2061.944625	0.028114	6.217536	608.933594
P	max	fast	30	2091.745296	0.043104	9.347642	608.933594
P	max	I	5	2008.939537	0.001454	0.084256	111.242188
P	max	I	10	2010.803477	0.002391	0.149249	111.242188
P	max	I	30	2020.844202	0.007442	0.437580	111.242188
P	max	II	5	2007.619648	0.000790	0.086033	111.242188
P	max	II	10	2008.846404	0.001407	0.177553	111.242188
P	max	II	30	2010.667220	0.002323	0.472028	111.242188
P	max	random	5	2007.392872	0.000676	4.597065	608.468750
P	max	random	10	2008.895815	0.001432	4.615860	608.468750
P	max	random	30	2014.793613	0.004398	4.970663	608.468750
U	min	fast	5	2012.219707	0.011879	5.634794	608.992188
U	min	fast	10	2004.715419	0.015823	6.360772	608.992188
U	min	fast	30	1980.341463	0.028636	9.520533	608.992188
U	min	I	5	2033.311955	0.000791	0.076333	111.242188
U	min	I	10	2031.095424	0.001956	0.142342	111.242188
U	min	I	30	2021.212449	0.007151	0.385082	111.242188
U	min	II	5	2033.332104	0.000780	0.057471	111.242188
U	min	II	10	2032.027747	0.001466	0.101866	111.242188
U	min	II	30	2025.756490	0.004762	0.325627	111.242188
U	min	random	5	2033.509981	0.000687	4.619808	607.171875
U	min	random	10	2032.397418	0.001271	4.638316	607.171875
U	min	random	30	2027.681036	0.003751	4.938497	607.171875

Task 4 reference ratios and overlaps

profile	pool	method	k	gain_ratio	jaccard
U	all	fast	2	1.000000	1.000000
U	all	fast	3	1.000000	1.000000
U	all	I	2	0.138732	0.000000
U	all	I	3	0.184832	0.000000
U	all	II	2	0.138994	0.000000
U	all	II	3	0.114333	0.000000
U	all	random	2	0.062405	0.000000
U	all	random	3	0.061638	0.000000
U	C	fast	2	1.000000	1.000000
U	C	fast	3	1.000000	1.000000
U	C	I	2	0.961558	0.333333
U	C	I	3	0.669833	0.500000
U	C	II	2	0.961558	0.333333
U	C	II	3	0.669833	0.500000
U	C	random	2	0.358003	0.050000
U	C	random	3	0.393214	0.095000
P	all	fast	2	1.000000	1.000000
P	all	fast	3	1.000000	1.000000
P	all	I	2	0.054229	0.000000
P	all	I	3	0.058168	0.000000
P	all	II	2	0.043117	0.000000
P	all	II	3	0.039782	0.000000
P	all	random	2	0.020930	0.000000
P	all	random	3	0.023430	0.000000
P	C	fast	2	1.000000	1.000000
P	C	fast	3	1.000000	1.000000
P	C	I	2	0.147184	0.000000
P	C	I	3	0.171581	0.000000
P	C	II	2	0.158482	0.000000
P	C	II	3	0.130407	0.000000
P	C	random	2	0.200566	0.050000
P	C	random	3	0.217275	0.040000

Task 4 greedy choices and bonus

Scatter uses innate s against the PRE-intervention mean neighbor equilibrium Pz0. The rank-one gain has positive column sum and denominator, so raising alpha helps maximisation when s exceeds the current neighbor average, and lowering helps when it is below. The pre-intervention explanatory plot can differ from current-round conditions.

Counts of consecutive marginal improvements increasing by more than 1e-8: {"U_max": 3, "P_max": 2, "U_min": 1}

Bonus scan order is ascending selected original ID, then the deterministic descending-degree shortlist (ID ties), with lower endpoint tested first. Exact equal-gain moves keep the first encountered move. Improvements must exceed 1e-8.

Bonus scans all flips and swaps of a selected target with an unselected member of the 200-degree shortlist, testing both inserted endpoints. Removal restores original alpha. Deterministic best improvement above 1e-8 repeats until a complete scan finds none. Each accepted move is independently checked. This is a local optimum for the specified neighborhood, not a global certificate.

U max: {"1": {"initial": 2038.6040132645596, "f": 2038.6040132645596, "added_gain": 0.0, "scans": 1, "evaluations": 399, "already_local": true, "seconds": 3.7794758770000954, "peak_mib": 607.7734375}, "2": {"initial": 2042.216701055877, "f": 2042.216701055877, "added_gain": 0.0, "scans": 1, "evaluations": 794, "already_local": true, "seconds": 4.782344379999813, "peak_mib": 624.50390625}, "3": {"initial": 2045.7202392502243, "f": 2045.7202392502243, "added_gain": 0.0, "scans": 1, "evaluations": 1185, "already_local": true, "seconds": 4.788798187999873, "peak_mib": 624.50390625}, "4": {"initial": 2049.122410616686, "f": 2049.122410616686, "added_gain": 0.0, "scans": 1, "evaluations": 1572, "already_local": true, "seconds": 4.276614193000114, "peak_mib": 624.50390625}, "5": {"initial": 2051.966520995197, "f": 2051.966520995197, "added_gain": 0.0, "scans": 1, "evaluations": 1965, "already_local": true, "seconds": 5.10139828299998, "peak_mib": 624.50390625}, "10": {"initial": 2061.5206915735303, "f": 2061.5206915735303, "added_gain": 0.0, "scans": 1, "evaluations": 3870, "already_local": true, "seconds": 3.7875530910000634, "peak_mib": 624.50390625}, "20": {"initial": 2075.305882221327, "f": 2075.305882221327, "added_gain": 0.0, "scans": 1, "evaluations": 7700, "already_local": true, "seconds": 6.808800263000194, "peak_mib": 624.50390625}, "30": {"initial": 2087.0601825201143, "f": 2087.0601825201143, "added_gain": 0.0, "scans": 1, "evaluations": 11370, "already_local": true, "seconds": 5.037962301999869, "peak_mib": 624.50390625}}

P max: {"1": {"initial": 2025.0002512534916, "f": 2025.0002512534916, "added_gain": 0.0, "scans": 1, "evaluations": 399, "already_local": true, "seconds": 4.438723921000019, "peak_mib": 607.3359375}, "2": {"initial": 2032.7265434012863, "f": 2032.7265434012863, "added_gain": 0.0, "scans": 1, "evaluations": 794, "already_local": true, "seconds": 5.597990325999945, "peak_mib": 623.1640625}, "3": {"initial": 2039.5144078197384, "f": 2039.5144078197384, "added_gain": 0.0, "scans": 1, "evaluations": 1185, "already_local": true, "seconds": 4.429514451000159, "peak_mib": 623.1640625}, "4": {"initial": 2044.3953879523417, "f": 2044.3953879523417, "added_gain": 0.0, "scans": 1, "evaluations": 1580, "already_local": true, "seconds": 4.864621564000117, "peak_mib": 623.1640625}, "5": {"initial": 2048.756721892898, "f": 2048.756721892898, "added_gain": 0.0, "scans": 1, "evaluations": 1965, "already_local": true, "seconds": 5.09904412800006, "peak_mib": 623.1640625}, "10": {"initial": 2061.944625316178, "f": 2061.944625316178, "added_gain": 0.0, "scans": 1, "evaluations": 3870, "already_local": true, "seconds": 4.58495220199984, "peak_mib": 623.1640625}, "20": {"initial": 2077.992467948686, "f": 2077.992467948686, "added_gain": 0.0, "scans": 1,

Task 4 greedy choices and bonus (continued)

"evaluations": 7660, "already_local": true, "seconds": 4.36622978000014, "peak_mib": 623.1640625},
"30": {"initial": 2091.7452957340224, "f": 2091.7452957340224, "added_gain": 0.0, "scans": 1,
"evaluations": 11490, "already_local": true, "seconds": 7.482120097000006, "peak_mib": 623.1640625}}

U min: {"1": {"initial": 2023.3033109935884, "f": 2023.3033109935884, "added_gain": -0.0, "scans":
1, "evaluations": 399, "already_local": true, "seconds": 5.305032359999586, "peak_mib": 607.515625},
"2": {"initial": 2018.9955712287883, "f": 2018.9955712287883, "added_gain": -0.0, "scans": 1,
"evaluations": 794, "already_local": true, "seconds": 4.565668920000007, "peak_mib": 623.2890625},
"3": {"initial": 2016.5004854344747, "f": 2016.5004854344747, "added_gain": -0.0, "scans": 1,
"evaluations": 1185, "already_local": true, "seconds": 3.355114937000053, "peak_mib": 623.2890625},
"4": {"initial": 2014.0386990292645, "f": 2014.0386990292645, "added_gain": -0.0, "scans": 1,
"evaluations": 1572, "already_local": true, "seconds": 3.854750901999978, "peak_mib": 623.2890625},
"5": {"initial": 2012.2197066035499, "f": 2012.2197066035499, "added_gain": -0.0, "scans": 1,
"evaluations": 1965, "already_local": true, "seconds": 5.639408872999866, "peak_mib": 623.2890625},
"10": {"initial": 2004.7154185312918, "f": 2004.7154185312918, "added_gain": -0.0, "scans": 1,
"evaluations": 3890, "already_local": true, "seconds": 9.672984551999889, "peak_mib": 623.3359375},
"20": {"initial": 1991.962672315301, "f": 1991.962672315301, "added_gain": -0.0, "scans": 1,
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"30": {"initial": 1980.3414625390528, "f": 1980.3414625390528, "added_gain": -0.0, "scans": 1,
"evaluations": 11370, "already_local": true, "seconds": 6.704584596999666, "peak_mib":
623.33984375}}

Bonus summary: U max

Greedy was already locally optimal in 8/8 tested budgets (100.0%).

Flips revise earlier endpoint choices after the surrounding equilibrium changes. Swaps let the algorithm undo an earlier node choice. These moves can escape the irreversible decisions of forward greedy; if no gain occurs, the completed neighborhood scan establishes local optimality up to 1e-8 for this neighborhood.

Baseline and Random comparisons, exact restricted references, and the bonus share identical initial parameters within each scenario. Raw outputs retain all target IDs, endpoint assignments and per-round values.

Bonus summary: P max

Greedy was already locally optimal in 8/8 tested budgets (100.0%).

Flips revise earlier endpoint choices after the surrounding equilibrium changes. Swaps let the algorithm undo an earlier node choice. These moves can escape the irreversible decisions of forward greedy; if no gain occurs, the completed neighborhood scan establishes local optimality up to 1e-8 for this neighborhood.

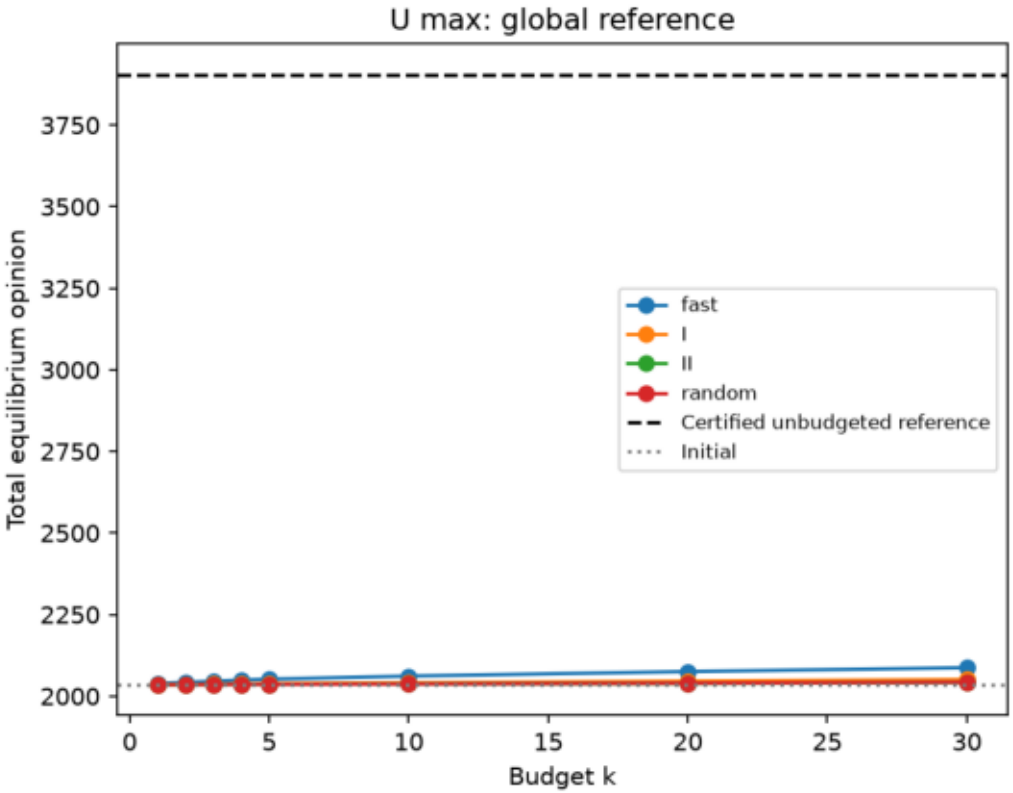
Baseline and Random comparisons, exact restricted references, and the bonus share identical initial parameters within each scenario. Raw outputs retain all target IDs, endpoint assignments and per-round values.

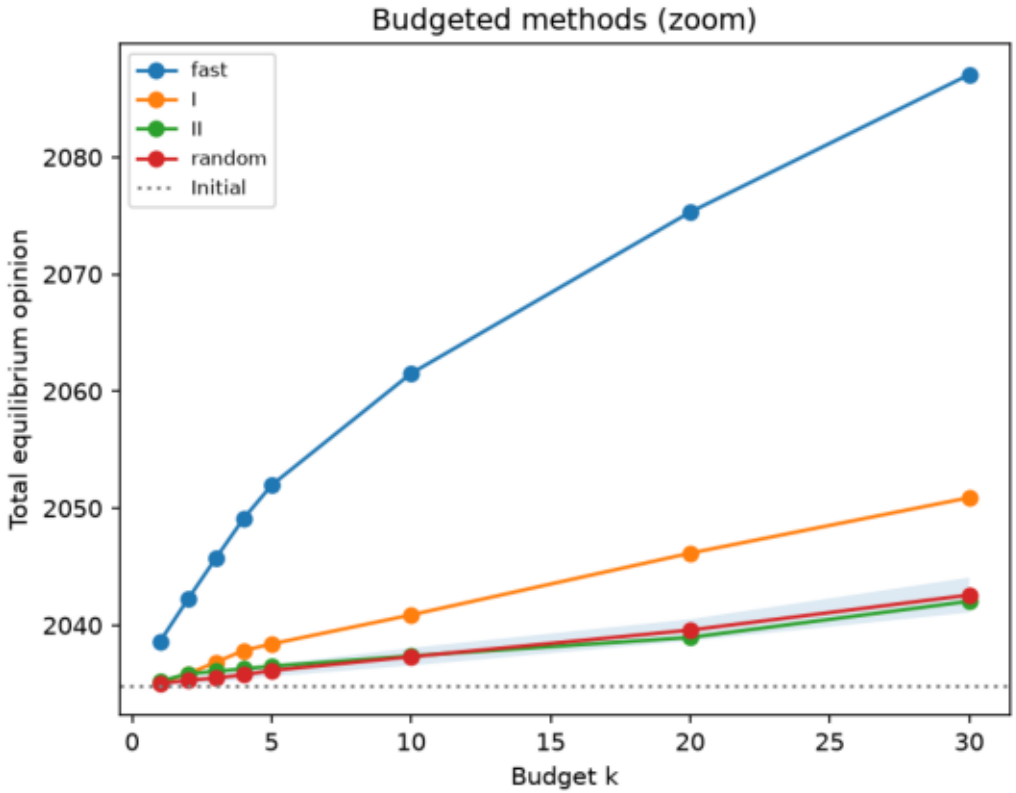
Bonus summary: U min

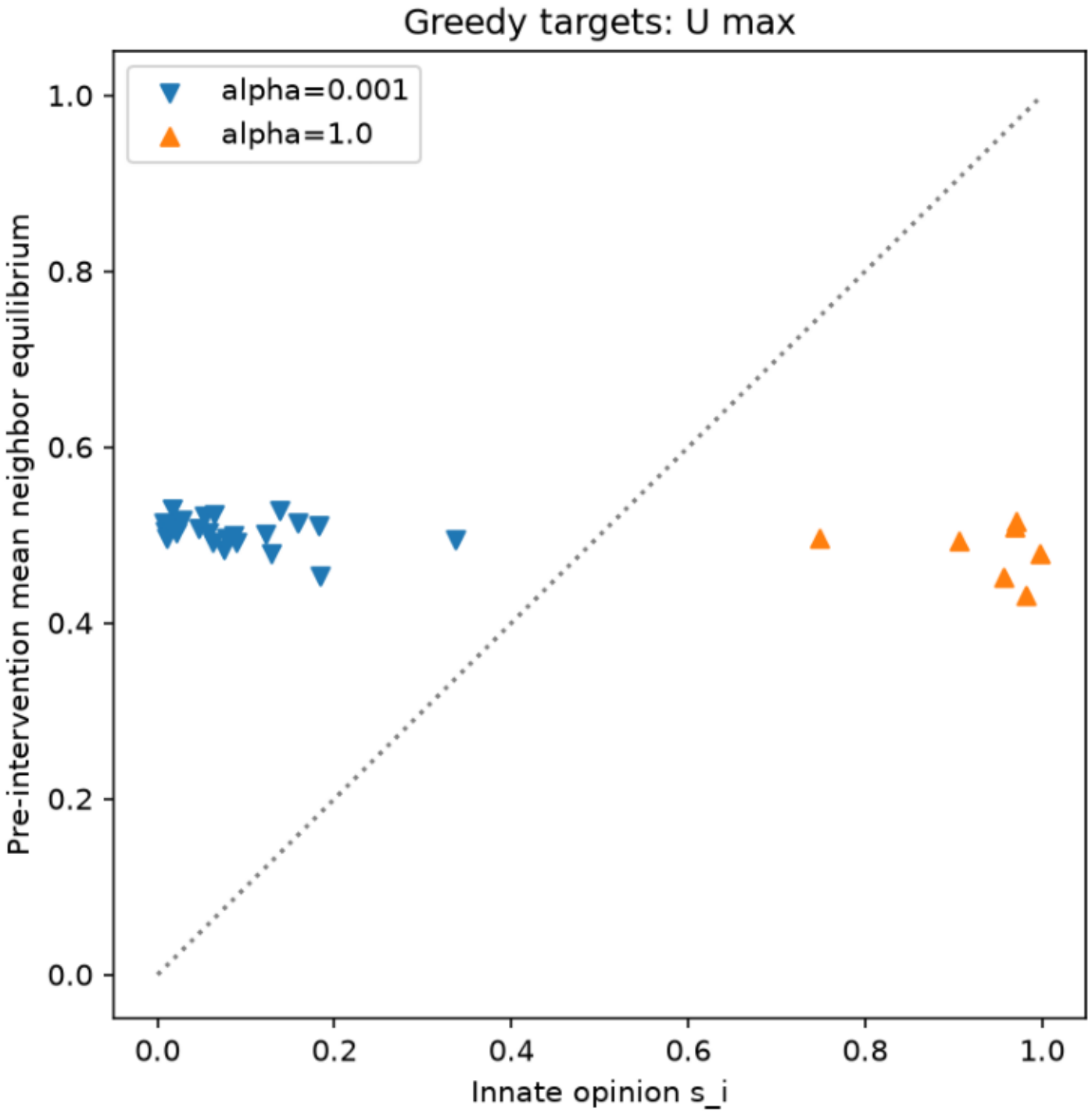
Greedy was already locally optimal in 8/8 tested budgets (100.0%).

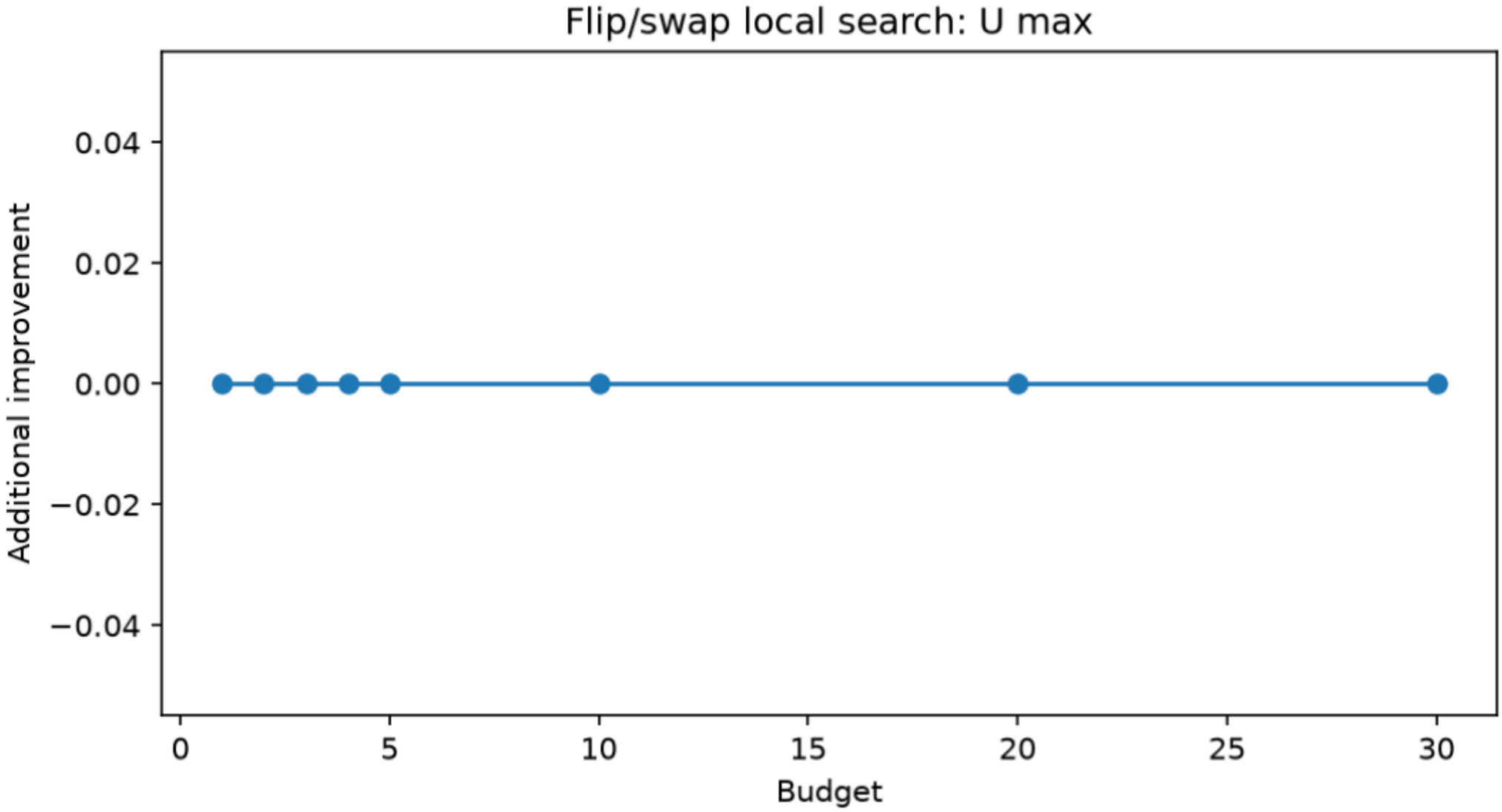
Flips revise earlier endpoint choices after the surrounding equilibrium changes. Swaps let the algorithm undo an earlier node choice. These moves can escape the irreversible decisions of forward greedy; if no gain occurs, the completed neighborhood scan establishes local optimality up to 1e-8 for this neighborhood.

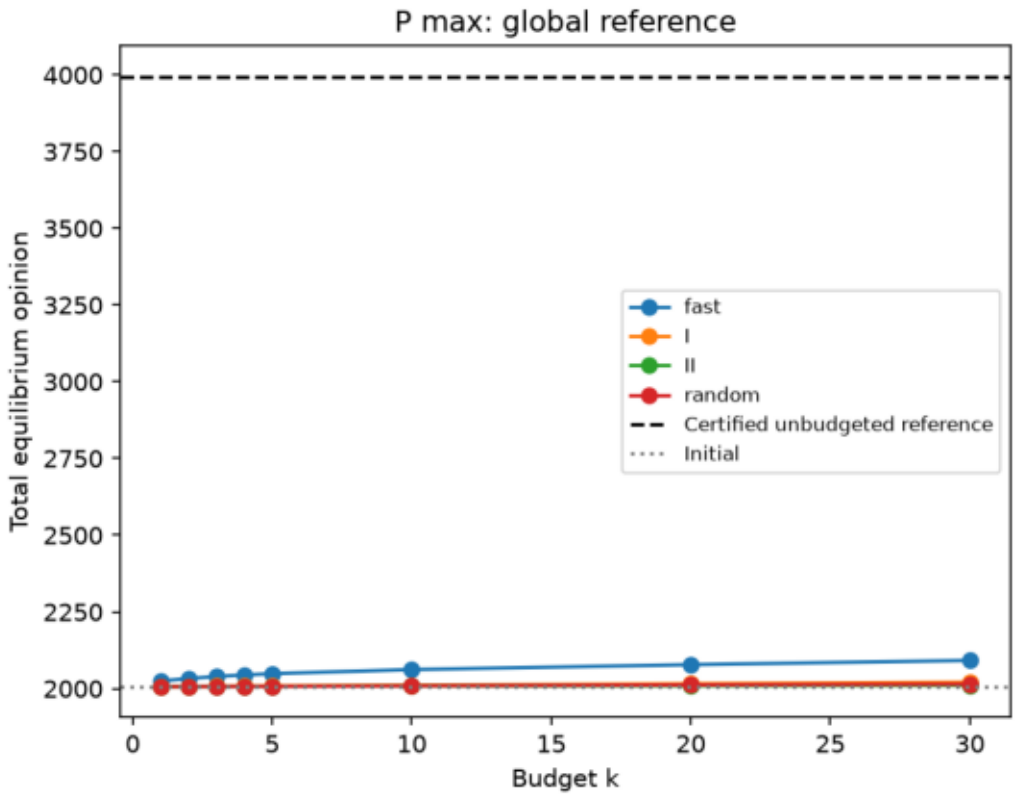
Baseline and Random comparisons, exact restricted references, and the bonus share identical initial parameters within each scenario. Raw outputs retain all target IDs, endpoint assignments and per-round values.

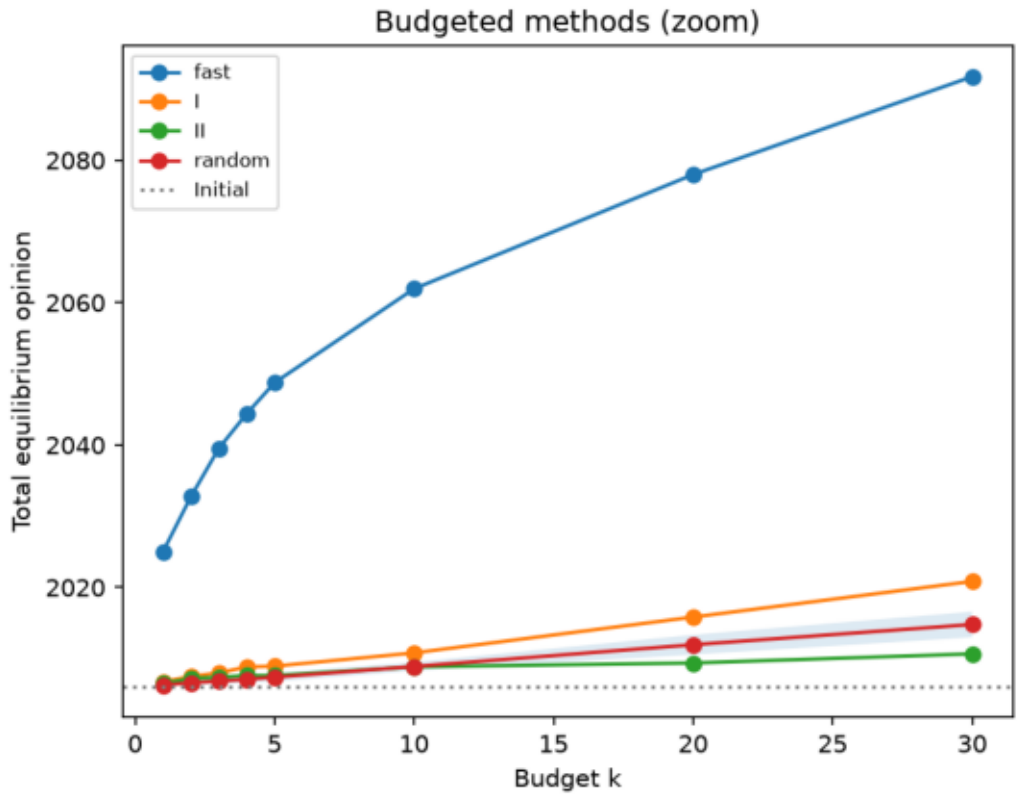

U max: global reference
Total equilibrium opinion
Budget k
fast
I
II
random
Certified unbudgeted reference
Initial
3750
3500
3250
3000
2750
2500
2250
2000
0
5
10
15
20
25
30

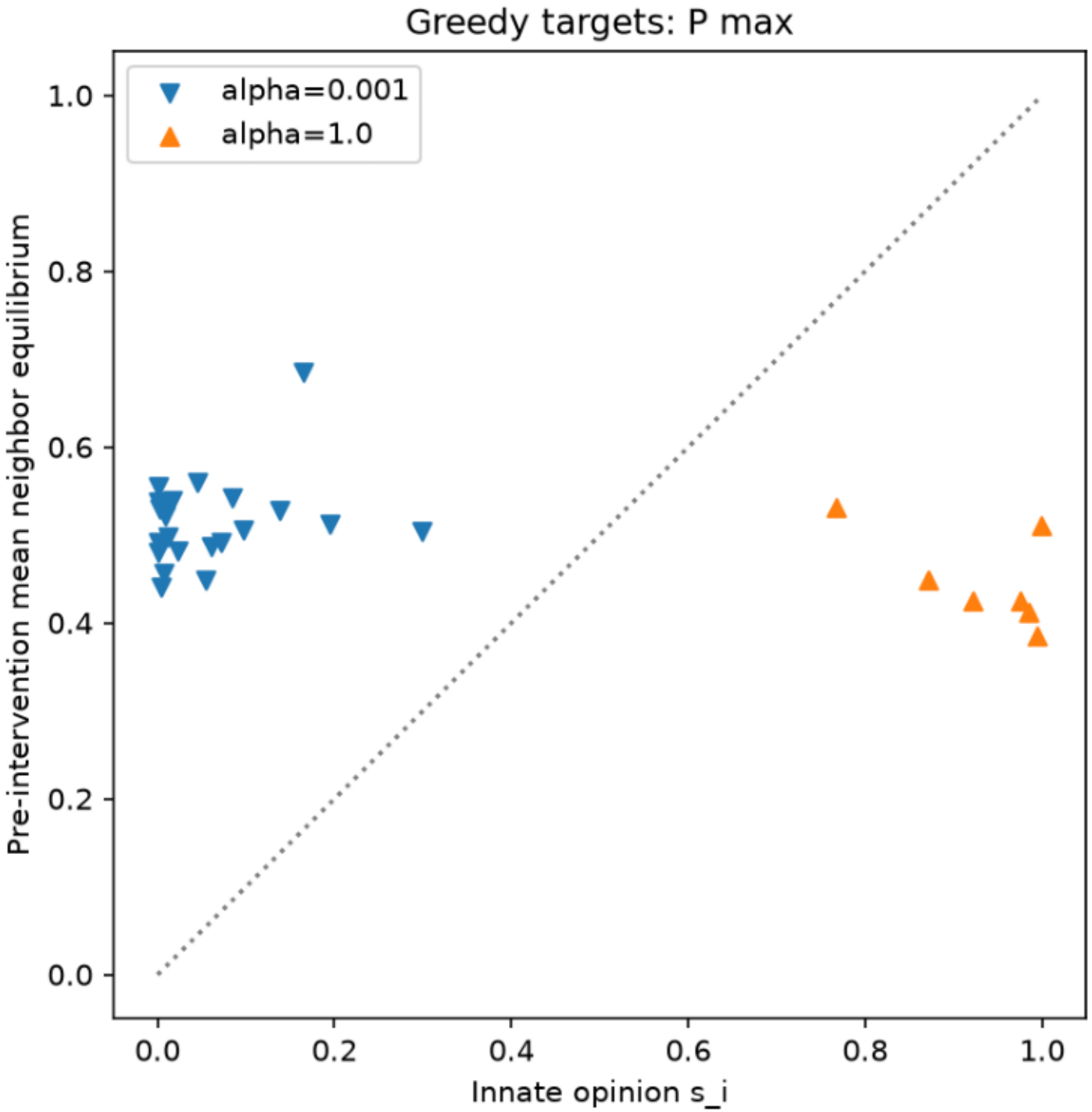

Budgeted methods (zoom)
Total equilibrium opinion
Budget k
fast
I
II
random
Initial
2080
2070
2060
2050
2040
0
5
10
15
20
25
30

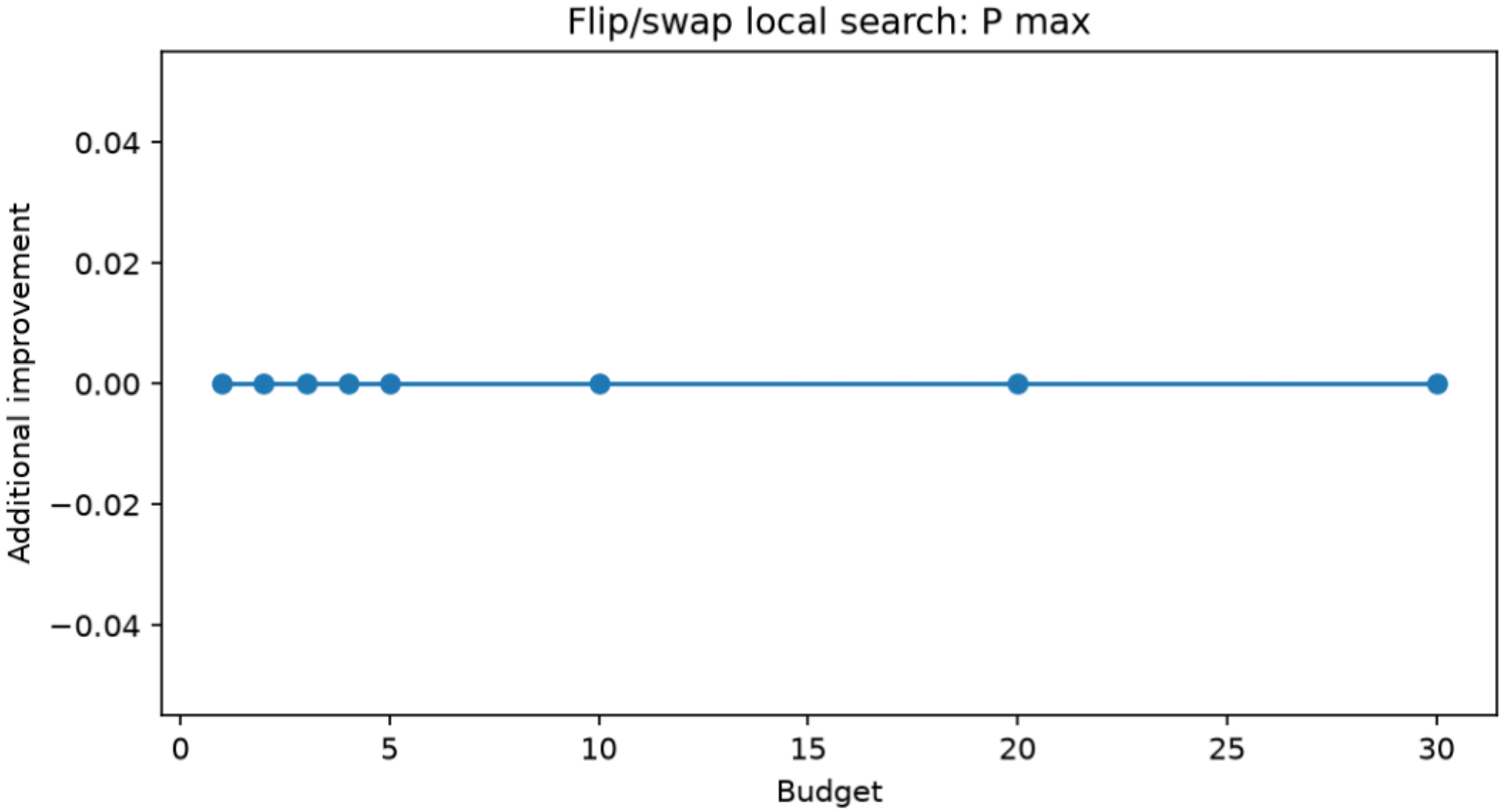

Greedy targets: U max
alpha=0.001
alpha=1.0
Pre-intervention mean neighbor equilibrium
Innate opinion s_i
0.0
0.2
0.4
0.6
0.8
1.0

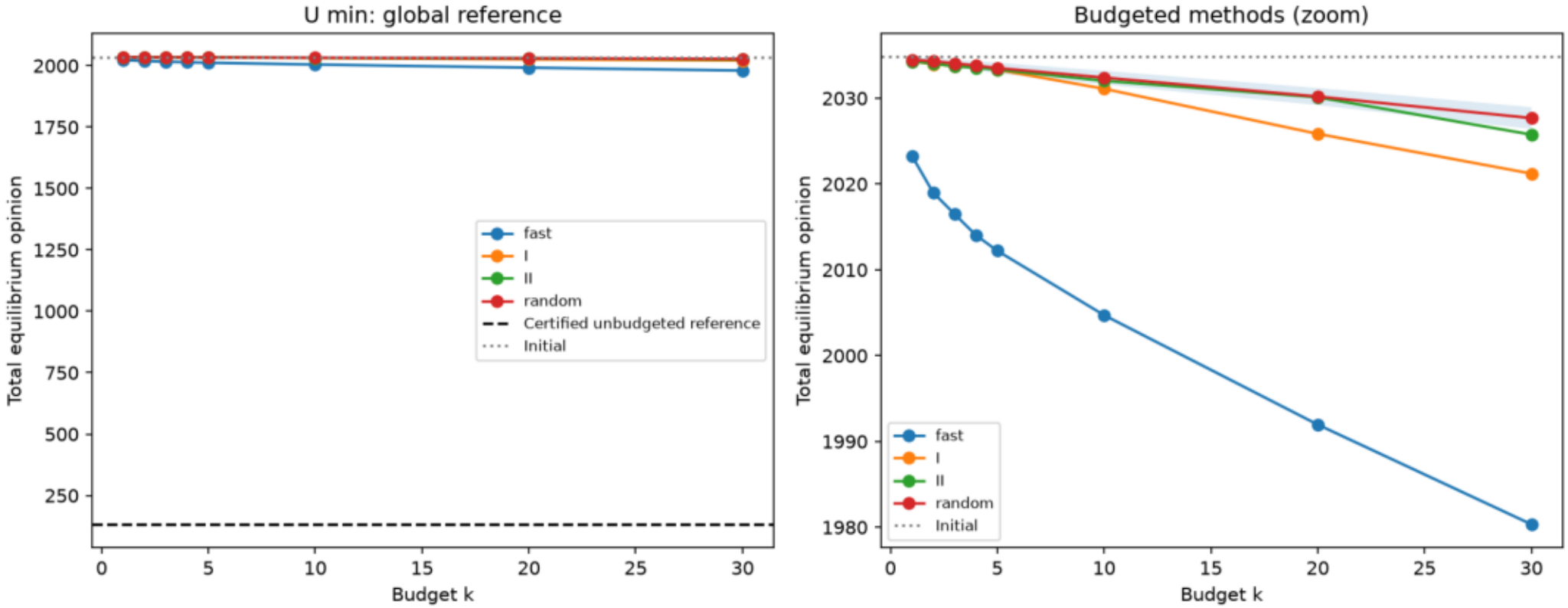

Flip/swap local search: U max
Additional improvement
0.04
0.02
0.00
−0.02
−0.04
0
5
10
15
20
25
30
Budget

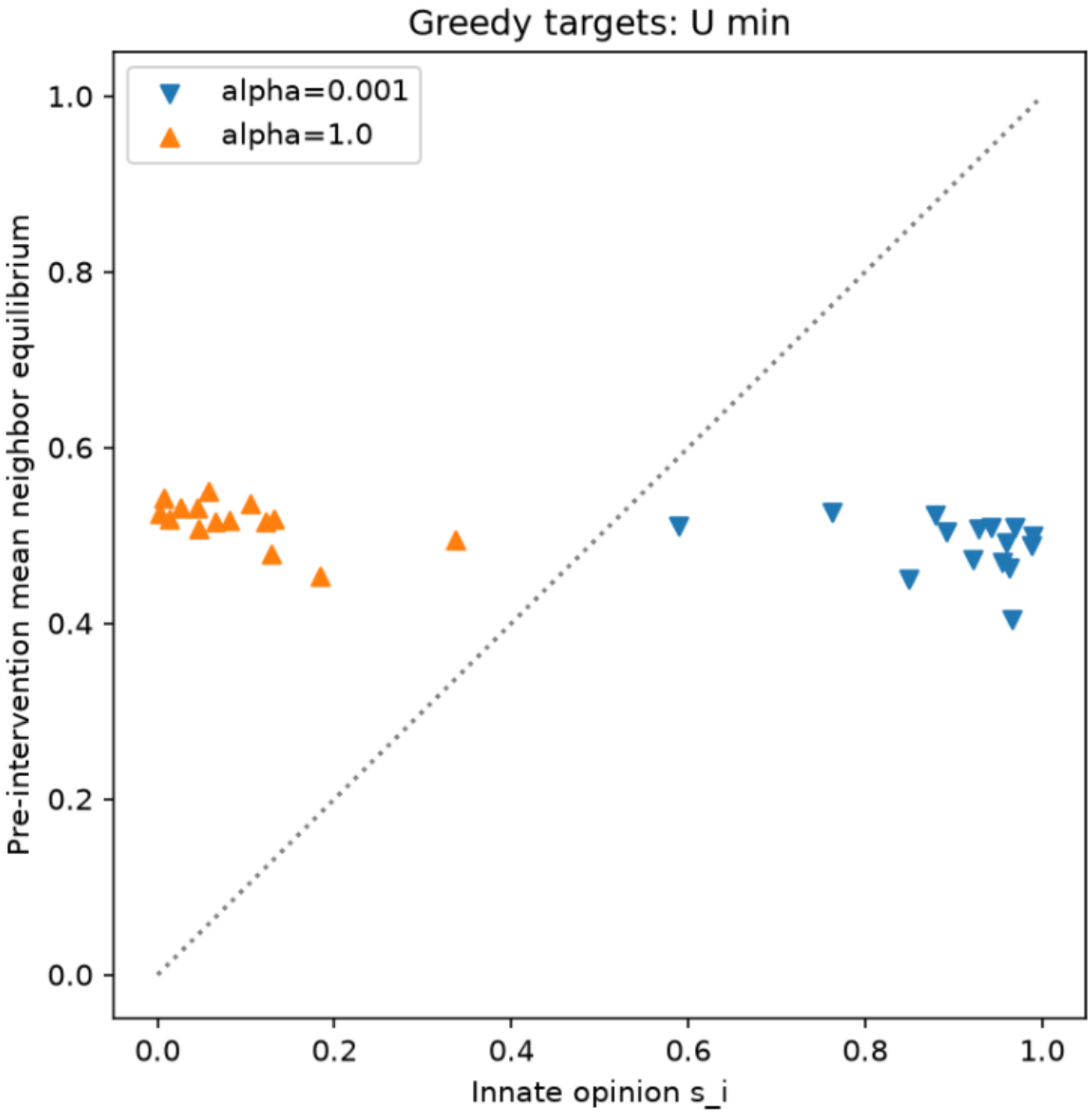

P max: global reference
Total equilibrium opinion
Budget k
fast
I
II
random
Certified unbudgeted reference
Initial
4000
3750
3500
3250
3000
2750
2500
2250
2000
0
5
10
15
20
25
30

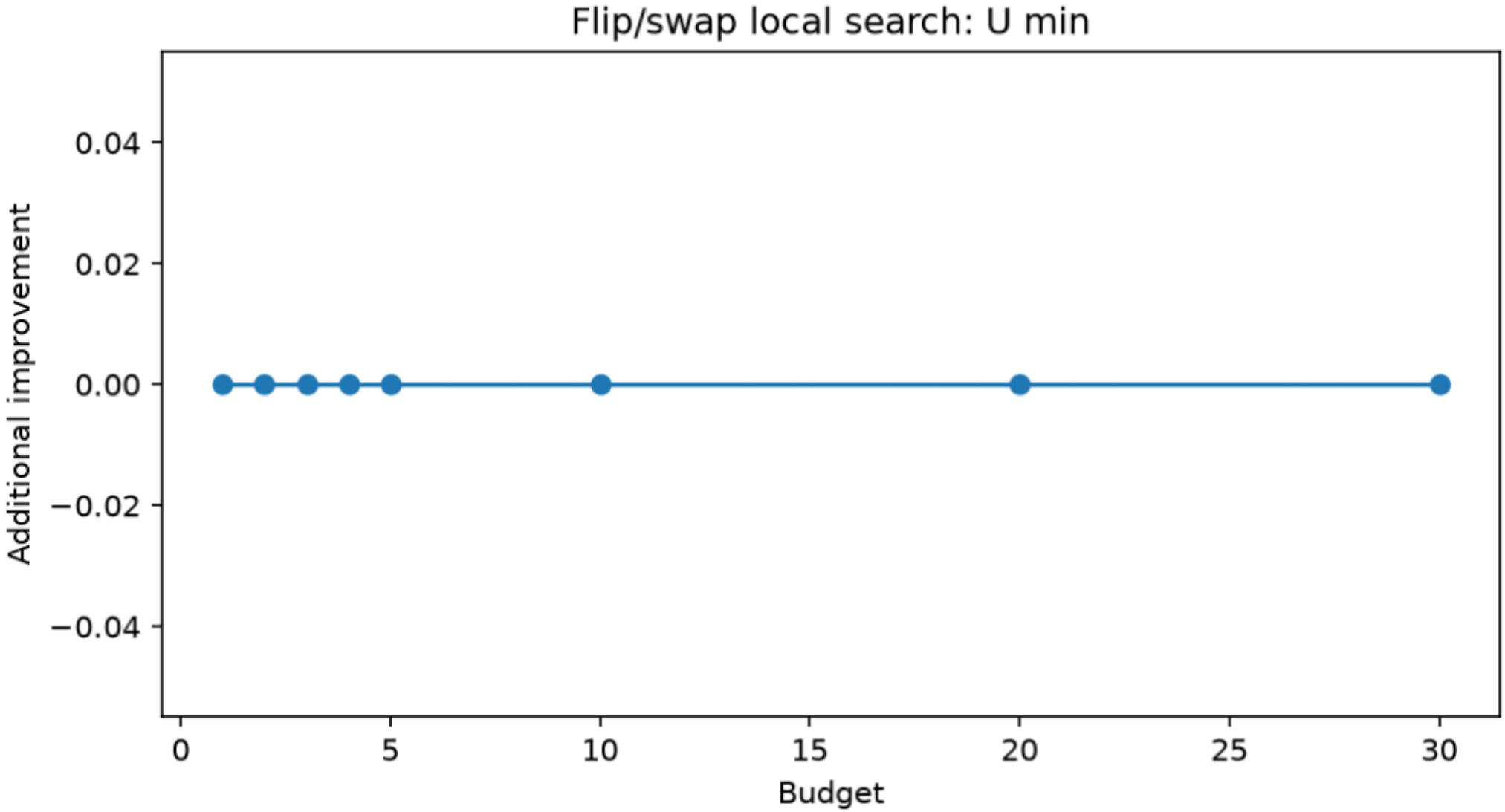

Budgeted methods (zoom)
Total equilibrium opinion
Budget k
fast
I
II
random
Initial
2080
2060
2040
2020
0
5
10
15
20
25
30

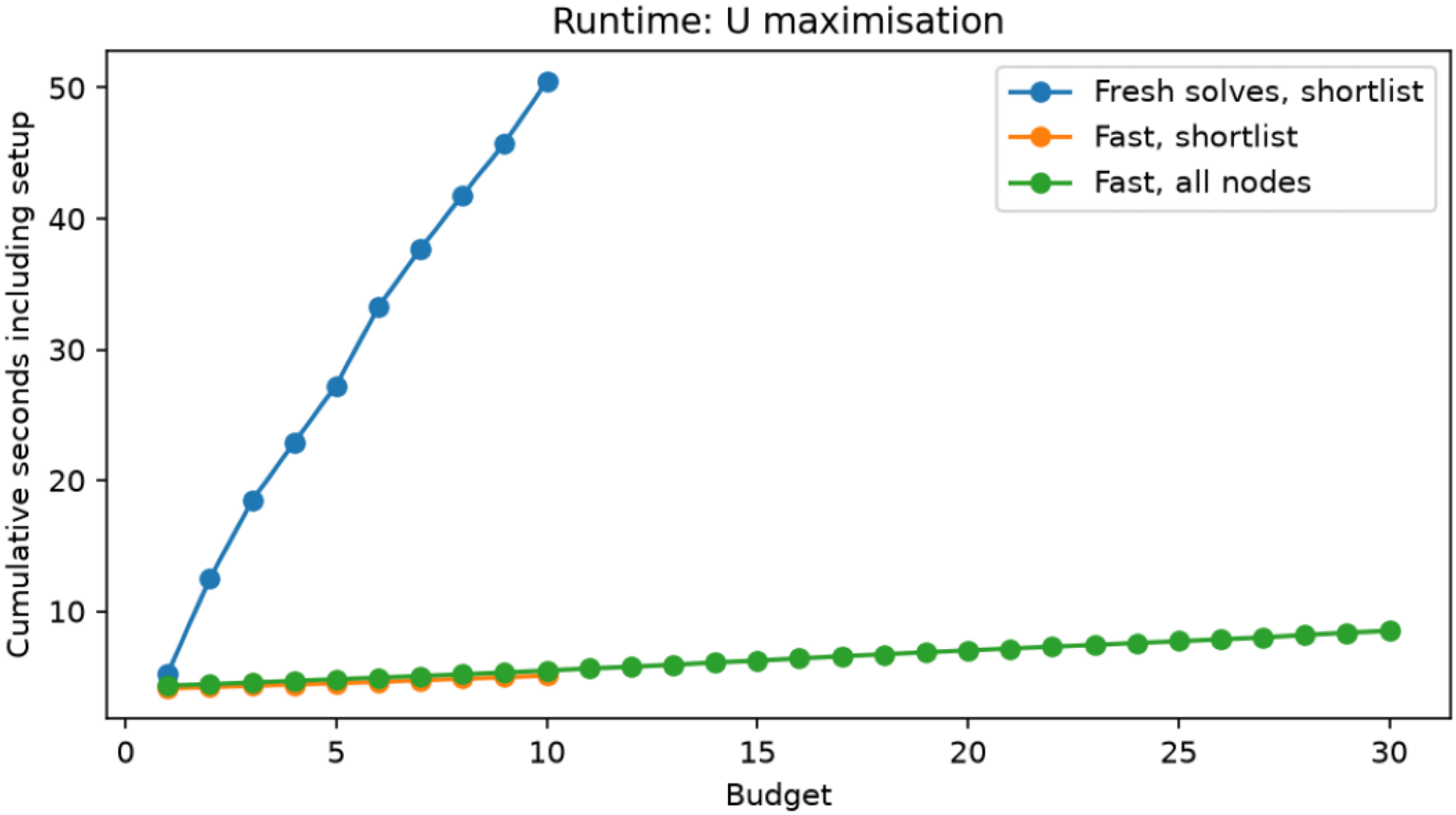

Greedy targets: P max
alpha=0.001
alpha=1.0
Pre-intervention mean neighbor equilibrium
Innate opinion s_i
0.0
0.2
0.4
0.6
0.8
1.0

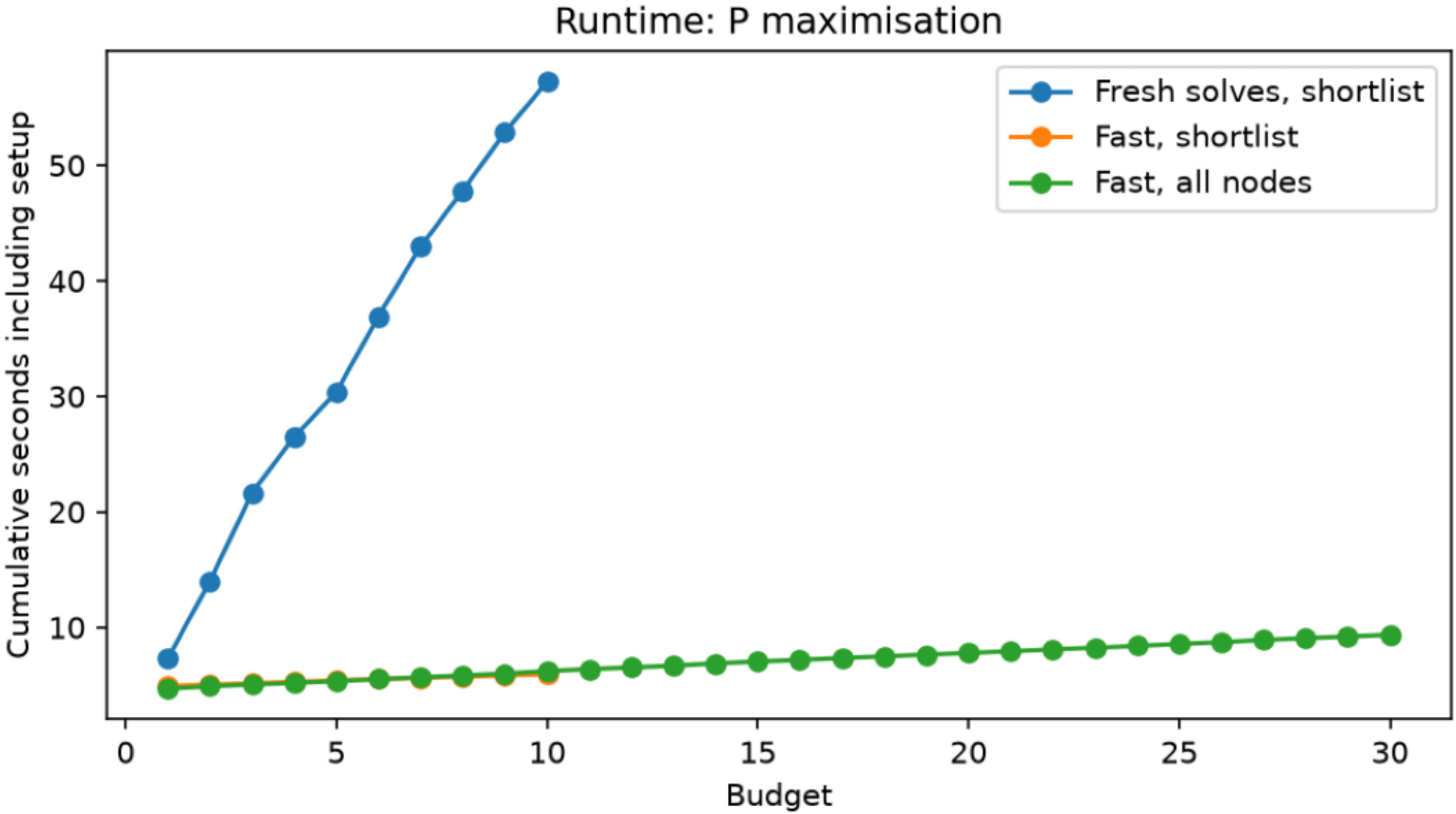

Flip/swap local search: P max
Additional improvement
0.04
0.02
0.00
−0.02
−0.04
0
5
10
15
20
25
30
Budget

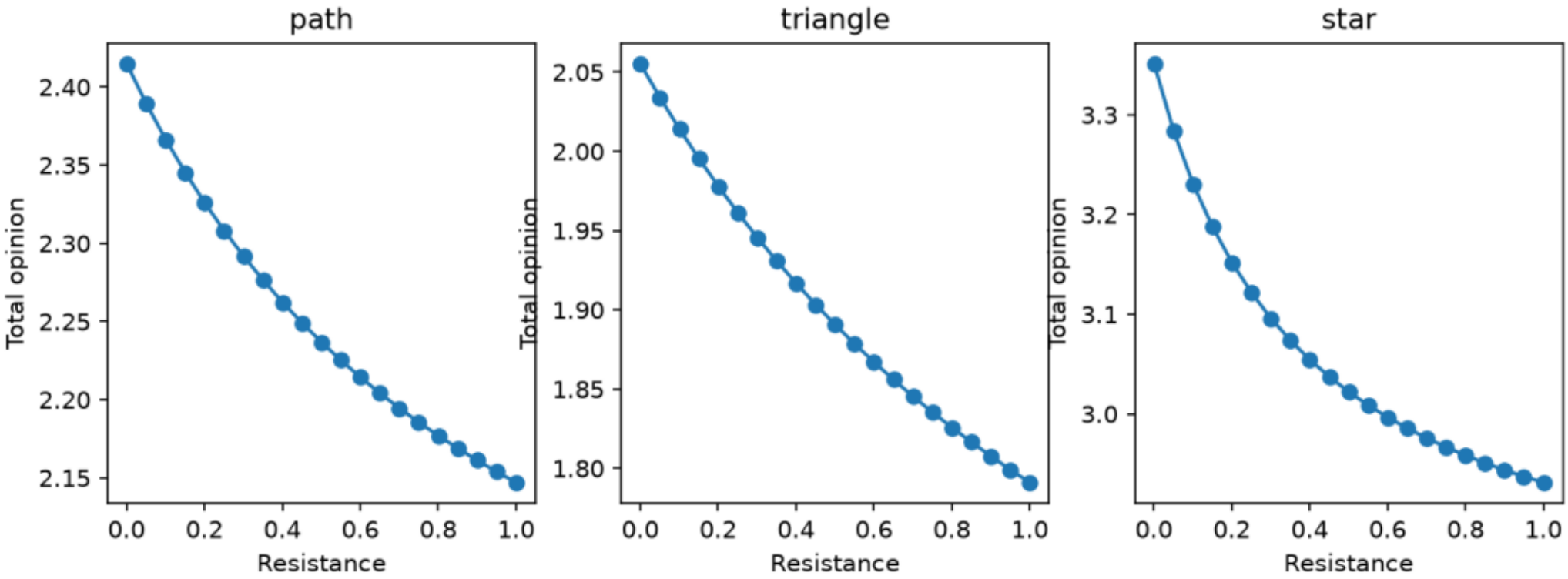

U min: global reference
Total equilibrium opinion
Budget k
fast
I
II
random
Certified unbudgeted reference
Initial
Budgeted methods (zoom)
Total equilibrium opinion
Budget k
fast
I
II
random
Initial


Greedy targets: U min
alpha=0.001
alpha=1.0
Pre-intervention mean neighbor equilibrium
Innate opinion s_i
0.0
0.2
0.4
0.6
0.8
1.0


Flip/swap local search: U min
Additional improvement
0.04
0.02
0.00
−0.02
−0.04
0
5
10
15
20
25
30
Budget


Runtime: U maximisation
Fresh solves, shortlist
Fast, shortlist
Fast, all nodes
Cumulative seconds including setup
50
40
30
20
10
0
5
10
15
20
25
30
Budget


Runtime: P maximisation
Fresh solves, shortlist
Fast, shortlist
Fast, all nodes
Cumulative seconds including setup
10
20
30
40
50
0
5
10
15
20
25
30
Budget


path
triangle
star
Total opinion
Resistance
2.40
2.35
2.30
2.25
2.20
2.15
2.05
2.00
1.95
1.90
1.85
1.80
3.3
3.2
3.1
3.0
0.0
0.2
0.4
0.6
0.8
1.0

Commands producing the numerical results

Commands below are the actual execution commands in the cloud workspace. For another location, use the included scripts with --dataset and --output pointing to the extracted input/results. The README gives portable commands.

```
/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_greedy_task1.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --profile U

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method fast --profile U --pool shortlist --k 10

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_greedy_task1.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --profile P

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method fast --profile P --pool shortlist --k 10

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method fast --profile U --direction max

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method I --profile U --direction max

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method II --profile U --direction max

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method random --profile U --direction max

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method fast --profile P --direction max

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method I --profile P --direction max

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method II --profile P --direction max
```

Commands producing the numerical results (continued)

```
/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method random --profile P --direction max

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method fast --profile U --direction min

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method I --profile U --direction min

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method II --profile U --direction min

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method random --profile U --direction min

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method pgd --profile U --direction max

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method pgd --profile U --direction min

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method witness --profile U

/workspace/soccomp/.venv/bin/python
/workspace/soccomp/asgn2/Assignment2_23CS10012_groundtruth_task3.py --dataset
/workspace/soccomp/asgn2/data/facebook_combined.txt --output /workspace/soccomp/asgn2/results
--profile U --restricted-comparisons

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method pgd --profile P --direction max

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method pgd --profile P --direction min

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_paper_task2.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --method witness --profile P
```

Commands producing the numerical results (continued)

```
/workspace/soccomp/.venv/bin/python
/workspace/soccomp/asgn2/Assignment2_23CS10012_groundtruth_task3.py --dataset
/workspace/soccomp/asgn2/data/facebook_combined.txt --output /workspace/soccomp/asgn2/results
--profile P --restricted-comparisons

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_bonus_task4.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --profile U --direction max

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_bonus_task4.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --profile P --direction max

/workspace/soccomp/.venv/bin/python /workspace/soccomp/asgn2/Assignment2_23CS10012_bonus_task4.py
--dataset /workspace/soccomp/asgn2/data/facebook_combined.txt --output
/workspace/soccomp/asgn2/results --profile U --direction min
```

References and reproducibility

Abebe, Kleinberg, Parkes, Tsourakakis. Opinion Dynamics with Varying Susceptibility to Persuasion. KDD 2018, pp.1089-1098. DOI:10.1145/3219819.3219983. Sections 2,4,5 supply endpoint, greedy and baseline definitions.

SNAP ego-Facebook: https://snap.stanford.edu/data/ego-Facebook.html

CS60017 Assignment: Opinion Maximisation on ego-Facebook, supplied four-page assignment, deadline 13 November 2026.

Abebe, Chan, Kleinberg, Liang, Parkes, Sozio, Tsourakakis. Opinion Dynamics with Varying Susceptibility to Persuasion via Non-Convex Local Search. arXiv:2011.02583v1, 4 November 2020. https://arxiv.org/abs/2011.02583. Abstract, introduction and Section 5 were inspected for the correction discussed here.

Raw per-round records, target IDs, assigned endpoints, seeds, reference certificates, command list, input arrays/checksums and software metadata accompany this report. Generated tables and figures derive exclusively from those completed experiments.